

UNESCO inaugurated the celebration of 2019 as the International Year of the Periodic Table of Chemical Elements to celebrate its completion of 150 years. The year 2019 marks the 150th anniversary of the creation of periodic table by Russian scientist Dmitry Ivanovich Mendeleev in 1869.

3. What is the usual property of oxides of Group 3 and 4?

- (a) Basic and Acidic (b) Basic
(c) Acidic (d) Neutral

43rd B.P.S.C. (Pre) 1999

Ans. (a)

The usual property of oxides of Group 3 and 4 of the periodic table are basic and acidic. These oxides are Amphoteric behave as both acidic and basic oxides.

4. The element found maximum in the soil layer is –

- (a) Oxygen (b) Nitrogen
(c) Manganese (d) Silicon

42nd B.P.S.C. (Pre) 1997

Ans. (a)

The element found the maximum in the soil layer is Oxygen 46.60% followed by Silicon 27.72%, Aluminium 8.13%, Iron 5.00% and Calcium 3.65%.

5. Which of the following is not a type of element?

- (a) Metals
(b) Non-metals
(c) Gases
(d) Metalloids
(e) None of the above/More than one of the above

65th B.P.S.C. (Pre) 2019

Ans. (e)

Elements of the periodic table are grouped as metals, metalloids (or semimetals) and nonmetals. Some Gases (hydrogen, helium, nitrogen, oxygen, fluorine, neon, chlorine, argon, krypton, xenon and radon) are elements & categorised as non-metals, but most of the gases are compounds.

6. Acids react with metal carbonates to liberate _____ gas.

- (a) H₂ (b) CO₂
(c) CO (d) More than one of the above
(e) None of the above

68th B.P.S.C. (Pre) 2022

Ans. (b)

When acids react with metal carbonates, they release carbon dioxides (CO₂) gas. The chemical reaction between an acid and a metal carbonate produces a salt, water and carbon dioxide. For example–
 $2\text{HCl} + \text{Na}_2\text{CO}_3 \rightarrow 2\text{NaCl} + \text{H}_2\text{O} + \text{CO}_2$

7. The process in which oxidation number decreases is known as

- (a) reduction
(b) oxidation
(c) redox reaction
(d) More than one of the above
(e) None of the above

B.P.S.C. School Teacher (Class 6-8) 10-12-2023

Ans. (a)

An increase in the oxidation number of the element in the given substance is termed as oxidation process, a decrease in the oxidation number of the element in the given substance is termed as reduction process and reactions which involve change in oxidation number of the interacting species is term as redox reaction.

8. The acid which can destroy the colour of acidic KMnO₄ is

- (a) CH₃COOH
(b) C₆H₈O₇
(c) CH₃CH₂COOH
(d) More than one of the above
(e) None of the above

B.P.S.C. School Teacher (Class 9-10) 08-12-2023

Ans. (b)

The acid which can destroy the colour of acidic potassium permanganate (KMnO₄) is Citric acid (C₆H₈O₇). It is an important metabolite in the pathway of all aerobic organisms. It has a role as a food acidity regulator, a chelator, an antimicrobial agent and a fundamental metabolite.

Metals, Minerals, Ores:

Properties, Uses

1. Which among the following is also known as white metal?

- (a) Nickel (b) Rhodium
(c) Platinum (d) Palladium
(e) None of the above/More than one of the above

67th B.P.S.C. (Re-Exam) (Pre) 2021

Ans. (e)

Basically white metal is an alloy of lead and lithium or other metals such as cadmium, bismuth and zinc, rhodium, palladium, platinum etc.

2. The metallurgical process in which a metal is obtained in a fused state is called

- (a) Smelting
- (b) Roasting
- (c) Calcination
- (d) Froth floatation
- (e) None of the above/More than one of the above

67th B.P.S.C. (Re-Exam) (Pre) 2021

Ans. (a)

Metal is obtained at high temperature by the reduction of ore in the process of smelting. The metal is obtained in a fused state in this process.

3. Which of the following is the electron configuration of a metallic element?

- (a) 2, 8
- (b) 2, 8, 7
- (c) 2, 8, 8
- (d) 2, 8, 8, 2

45th B.P.S.C. (Pre) 2001

Ans. (d)

Calcium is a dull, grey, solid element with a silver appearance which exists in the solid state. It has a high melting point (1115 K) and boiling point (1757 K). All these features make it related to the metals. The electronic configuration of Calcium is 2,8,8,2. Hence, it has a tendency to lose two electrons to get a noble gas configuration. Since it can lose electrons, it can be used in ionic bonding and can form ionic compounds. Like other metals, Calcium also reacts vigorously with dilute acids like hydrochloric acid and produce large amounts of heat, forms Calcium Chloride (CaCl₂) and Hydrogen gas. All these properties of Calcium prove that it is a metal.

4. Which is the most reactive metal?

- (a) Sodium
- (b) Calcium
- (c) Iron
- (d) Potassium

56th to 59th B.P.S.C. (Pre) 2015

Ans. (d)

Among the given options, Potassium is the most reactive metal. The given metals in order of reactivity are- Potassium (K) > Sodium (Na) > Calcium (Ca) > Iron (Fe).

5. The cause of diamond shine is

- (a) size
- (b) cutting
- (c) reflection
- (d) total internal reflection
- (e) None of the above / More than one of the above

B.P.S.C. Auditor 2021

Ans. (d)

Diamonds are known for their spectacular brilliance. Their brilliance is mainly due to the total internal reflection of light inside them. The critical angle for diamond-air interface ($\cong 24.4^\circ$) is very small, therefore once light enters a diamond,

it is very likely to undergo total internal reflection inside it. Diamonds found in nature rarely exhibit the brilliance for which they are known. It is the technical skill of a diamond cutter which makes diamonds to sparkle so brilliantly. By cutting the diamond suitably, multiple total internal reflections can be made to occur.

6. Which is the hardest in the following?

- (a) Diamond
- (b) Glass
- (c) Quartz
- (d) Platinum

44th B.P.S.C. (Pre) 2000

Ans. (a)

Diamond is the hardest, least compressible and best thermal conductor among all natural materials.

7. Which of the following statements about graphite and diamond is true?

- (a) They have the same crystal structure.
- (b) They have the same hardness.
- (c) They have the same electrical conductivity.
- (d) They can undergo a similar chemical reaction.
- (e) None of the above / more than one of the above

67th B.P.S.C. (Pre) 2021

Ans. (d)

Both graphite and diamond are allotropes of carbon, but both have different crystalline structures. While diamond is the most naturally occurring carbon material, graphite has the least carbon. The electrical conductivity of graphite is higher than that of diamond. Since both graphite and diamond are made of covalent carbon bonds, they have similar chemical properties and can undergo a similar chemical process.

8. Which of the following statements about diamond and graphite is true?

- (a) They have same electrical conductivity.
- (b) They have same crystal structure.
- (c) They have same degree of hardness.
- (d) More than one of the above
- (e) None of the above

68th B.P.S.C. (Pre) 2022

Ans. (e)

Graphite and diamonds are both allotropes of carbon, which means they are composed of the same element but have different crystal structure and properties.

Electrical Conductivity—Graphite is an excellent electrical conductor, it has a high mobility due to the presence of delocalized electrons in its layered structure. Unlike Graphite, diamonds are poor electrical conductors.

Crystal Structure—Graphite has a layered crystal structure, while diamonds have a three-dimensional crystal structure known as diamond lattice or diamond cubic structure.

Degree of hardness—Graphite is a soft material with a low degree of hardness while diamonds are one of the hardest natural materials known.

9. The purest form of iron is

- (a) steel
- (b) pig iron
- (c) cast iron
- (d) wrought iron
- (e) None of the above/More than one of the above

B.P.S.C. CDPO 2018

Ans. (d)

Wrought iron is a soft, ductile, fibrous variety that is produced from a semifused mass of relatively pure iron globules partially surrounding by slag. It usually contains less than 0.1 percent carbon and 1 or 2 percent slag.

10. Heaviest metal of the following is made of –

- (a) Copper
- (b) Uranium
- (c) Aluminium
- (d) Silver

40th B.P.S.C. (Pre) 1995

Ans. (b)

Uranium is the heaviest naturally occurring element with an atomic no. of 92. It is a pure form of silver-coloured heavy metal. Its most common isotope Uranium-238 has a nucleus containing 92 protons and 146 neutrons. It has a density of 19 (Source–energy.gov site).

11. Consider the following statement with respect to noble metals:

- 1. Noble metals are found in pure form in nature.
- 2. Uranium and lead are example of noble metal.

Which of the above statement is/are correct?

- (a) 1 only
- (b) 2 only
- (c) Both 1 and 2
- (d) Brass is also noble metal
- (e) None of the above/More than one of the above

67th B.P.S.C. (Re-Exam) (Pre) 2021

Ans. (e)

Noble metal are found in nature in pure as well as mixed form. Uranium and lead are not examples of noble metals.

12. The heaviest metal among the following is :

- (a) Gold
- (b) Silver

- (c) Mercury
- (d) Platinum
- (e) None of the above/More than one of the above

63rd B.P.S.C. (Pre) 2017

Ans. (d)

Among the given options, platinum is the heaviest metal with a density of 21.45 g/cm³.

13. Which of the following metals does not corrode?

- (a) Gold
- (b) Iron
- (c) Platinum
- (d) More than one of the above

BPSC Agriculture Department-2024

Ans. (d)

Both gold and platinum are metals that do not corrode easily. Gold and platinum are stable metals that are resistant to oxygen and acids. They are also less reactive than other metals, so they don't react easily with other substances.

14. The chemical formula of sapphire (Ruby) is :

- (a) Al₂O₃
- (b) Al₃O₂
- (c) N₂O
- (d) NO₂
- (e) None of the above/More than one of the above

66th B.P.S.C. (Pre) (Re. Exam) 2020

Ans. (a)

The chemical formula of sapphire (Ruby) is Al₂O₃ (Aluminium Oxide). It is a precious gemstone, a variety of mineral corundum.

15. Al₂O₃.2SiO₂.2H₂O is the chemical formula for

- (a) monazite
- (b) china clay
- (c) sand
- (d) More than one of the above
- (e) None of the above

B.P.S.C. School Teacher (Class 6-8) 09-12-2023

Ans. (b)

Al₂O₃.2SiO₂.2H₂O is a chemical formula of china clay which is also known as Kaolin, softwhite clay that is an essential ingredient in the ceramic industry, especially in fine porcelains, because they can be easily molded have a fine texture.

16. A layer of green coloured basic carbonate gets deposited on a metal after keeping it in air for some time, that metal is :

- (a) Copper
- (b) Silver
- (c) Nickel
- (d) Zinc

46th B.P.S.C. (Pre) 2004

Ans. (a)

After keeping it in air for some time, a layer of green basic carbonate gets deposited on the copper metal. In fact, it is the result of oxidation of copper in air. It is called 'Patina'.

17. Gold is dissolved in –

- (a) Sulphuric acid
- (b) Nitric acid
- (c) Mixture of Sulphuric and Nitric Acid
- (d) Hydrochloric acid

47th B.P.S.C. (Pre) 2005

Ans. (*)

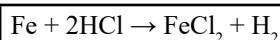
Gold is unaffected in air, water, alkali halogen and all acids except Aqua regia (a mixture of hydrochloric acid and nitric acid in a 3:1 ratio). The name Aqua-regia was coined by chemists because of its ability to dissolve gold "the king of metals". It is a mixture of acids, a fuming yellow or red solution.

18. What happens when dilute hydrochloric acid is added to iron filings?

- (a) Hydrogen gas and iron chloride are produced
- (b) Chlorine gas and iron hydroxide are produced
- (c) No reaction takes place
- (d) More than one of the above
- (e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (a)



When dilute hydrochloric acid (HCl) is added to iron (Fe) filing, hydrogen gas (H) and iron chloride (FeCl₂) are produced. In this reaction, iron displaces hydrogen from hydrochloric acid to form iron chloride and hydrogen gas.

19. Minerals are-

- (a) Liquids
- (b) Inorganic solids
- (c) Gases
- (d) All of the above

53rd to 55th B.P.S.C. (Pre) 2011

Ans. (b)

According to Geologists, minerals are naturally occurring inorganic substances with a definite and predictable chemical composition and physical properties.

20. Which of the following is the main ingredient of cement?

- (a) Limestone
- (b) Silica clay
- (c) Gypsum
- (d) Ash
- (e) None of the above / More than one of the above

67th B.P.S.C. (Pre) 2021

Ans. (e)

The main components used in the manufacture of cement are limestone and silica clay (SiO₂). When limestone and silica clay are heated together, their reaction results in the formation of cement clinker. Cement is formed when 2-3% gypsum (CaSO₄.2H₂O) is mixed in this clinker.

21. Iron is obtained from which of the following minerals?

- (a) Limestone
- (b) Pitchblende
- (c) Monazite
- (d) Hematite

42nd B.P.S.C. (Pre) 1997-98

Ans. (d)

Iron is obtained from hematite ore.

22. To protect broken bones, Plaster of Paris is used. It is :

- (a) Slaked lime
- (b) Calcium carbonate
- (c) Calcium oxide
- (d) Gypsum
- (e) None of the above/More than one of the above

63rd B.P.S.C. (Pre) 2017

Ans. (d)

Plaster of Paris is a white powdery slightly hydrated calcium sulfate (CaSO₄.1/2 H₂O) made by calcining gypsum and used mainly for casts and molds in the form of a quick-setting paste with water.

23. The chemical formula of the Plaster of Paris is –

- (a) CaSO₄
- (b) CaSO₄. ½ H₂O
- (c) CaSO₄.H₂O
- (d) CaSO₄.2H₂O

42nd B.P.S.C. (Pre) 1997

39th B.P.S.C. (Pre) 1994

Ans. (b)

See the explanation of the above question.

24. Which substance is formed by the action of chlorine gas on dry slaked lime?

- (a) Calcium chloride
- (b) Bleaching powder
- (c) Hydrochloric acid
- (d) More than one of the above
- (e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (b)

Bleaching powder is formed by the action of Chlorine gas on dry slaked lime. It is used as a source of Chlorine for various purposes. Its chemical name is Calcium Oxychloride and its chemical formula is CaOCl₂.

25. Which among the following is known as quicklime?

- (a) CaO
- (b) CaCO₂

- (c) $\text{Ca}(\text{OH})_2$
 (d) CaCl_2
 (e) None of the above/More than one of the above

67th B.P.S.C. (Re-Exam) (Pre) 2021

Ans. (a)

Calcium oxide (CaO) is called quicklime. It is a crystalline solid at room temperature.

26. The poorest conductor of heat among the following is :

- (a) Copper (b) Lead
 (c) Mercury (d) Zinc
 (e) None of the above / More than one of the above

66th B.P.S.C. (Pre) 2020

Ans. (c)

All metals are conductors of heat but lead and mercury are poor conductors of heat compared to other metals. Thermal conductivity of lead is about 34.7 w/mK while thermal conductivity of mercury is only about 8.31 w/mK. Hence, mercury is the poorest conductor of heat among all metals.

27. Which one of the following metals is liquid at ordinary temperature?

- (a) Lead (b) Nickel
 (c) Mercury (d) Tin

44th B.P.S.C. (Pre) 2000

Ans. (c)

Mercury is the only common metal which is liquid at ordinary (room) temperature. Mercury is sometimes called quicksilver.

28. There is no reaction when steam passes over –

- (a) Aluminium (b) Copper
 (c) Carbon (d) Iron

38th B.P.S.C. (Pre) 1992

Ans. (b)

Metals like potassium and sodium react violently with cold water. The reaction of calcium with water is less violent. Magnesium does not react with cold water but it reacts with hot water. Metals like aluminium, iron and zinc do not react either with cold or hot water but they react with steam. Metals such as lead, copper, silver, gold and platinum do not react with water at all (neither with cold nor hot water and not with steam).

29. Which of the following has highest melting point?

- (a) Boron (b) Iron
 (c) Silicon (d) Aluminium
 (e) None of the above/More than one of the above

66th B.P.S.C. (Pre) (Re. Exam) 2020

Ans. (a)

Melting point of Boron - 2348 K, Iron - 1811 K, Silicon - 1687 K and Aluminium - 933 K. Hence, among the given options, Boron has highest melting point.

30. Consider the following statements :

1. The Addition of salt to pure water increases the boiling point of water and decreases the freezing point of water.
2. When methyl alcohol is added to water, boiling point of water decreases.

Which of the above statements is/are correct?

- (a) Salt and methyl alcohol are same in behaviour
 (b) 1 only
 (c) 2 only
 (d) Both 1 and 2
 (e) None of the above/More than one of the above

67th B.P.S.C. (Re-Exam) (Pre) 2021

Ans. (d)

When the salt is dissolved in pure water, it increases the boiling point of water and decreases the freezing point of water. When methyl alcohol is added to water, it decreases the boiling point of water. This happens due to Presence hydrogen bonding.

31. What happens to the weight of iron, when it rusts?

- (a) Increases for long time (b) Decreases then increases
 (c) Increases then decreases (d) Remains same
 (e) None of the above/More than one of the above

67th B.P.S.C. (Re-Exam) (Pre) 2021

Ans. (e)

When iron oxidizes oxygen from air in presence of moisture it forms Iron oxide Ferrous oxide (Fe_2O_3) which is known as rust with the oxidation of Iron or the weight of iron will increase but after a long time the iron starts decaying due to rust. This will reduce the weight of Iron. Hence option (c) and (a) both is correct the correct answer for this question is option (e).

32. Chemically rust is

- (a) Hydrated ferric oxide (b) Hydrated ferrous oxide
 (c) Only ferric oxide (d) More than one of the above
 (e) None of the above

BPSC (Tre-3) {Class 1-5}–2024

Ans. (a)

Chemically, rust is primarily composed of iron oxides, which form when iron reacts with oxygen and moisture in the environment. The most common form of rust is hydrated iron oxide, represented by the formula $\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}$. This process is known as oxidation, and it typically occurs in the presence of water and electrolytes, which facilitate the transfer of electrons.